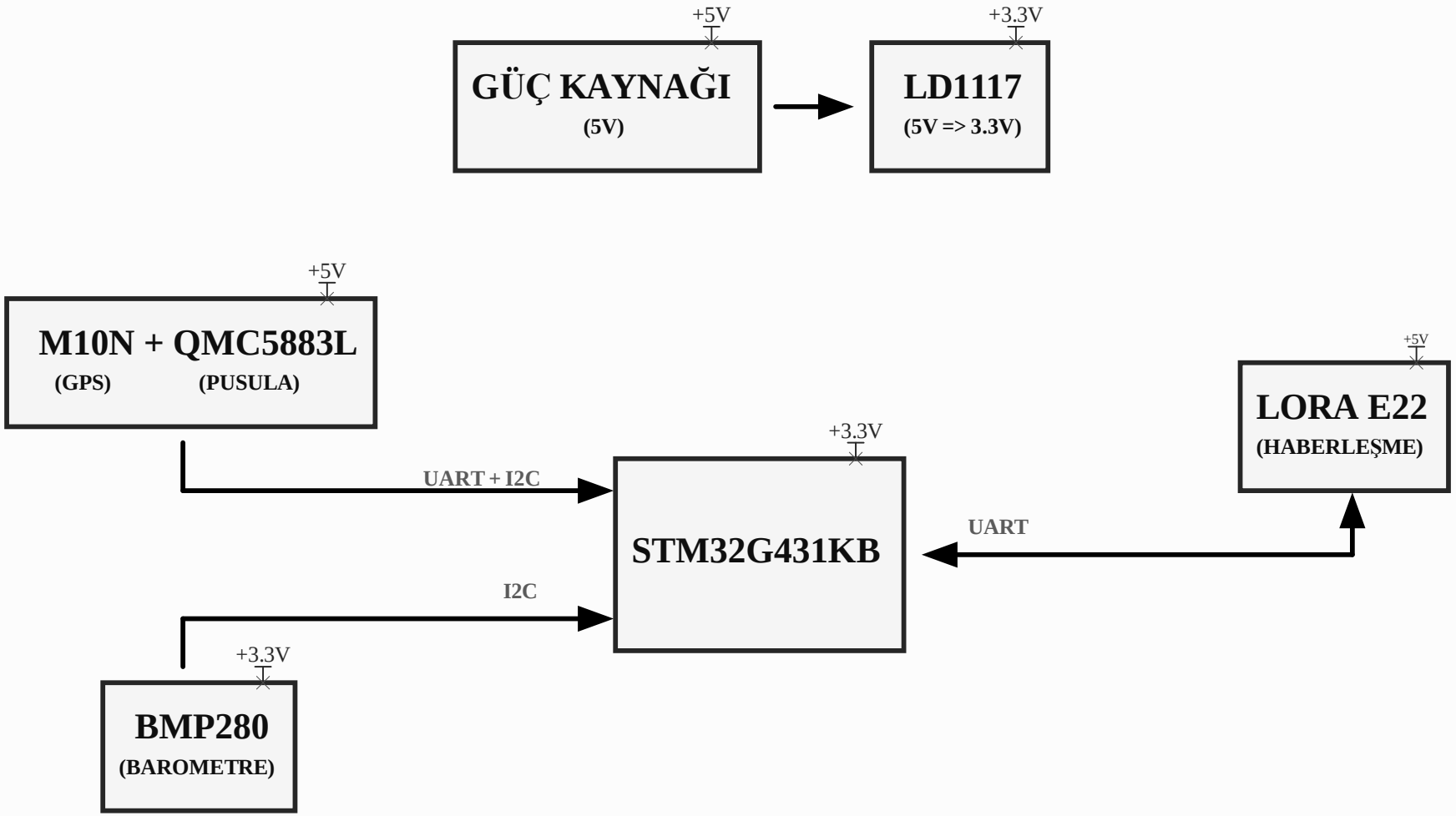
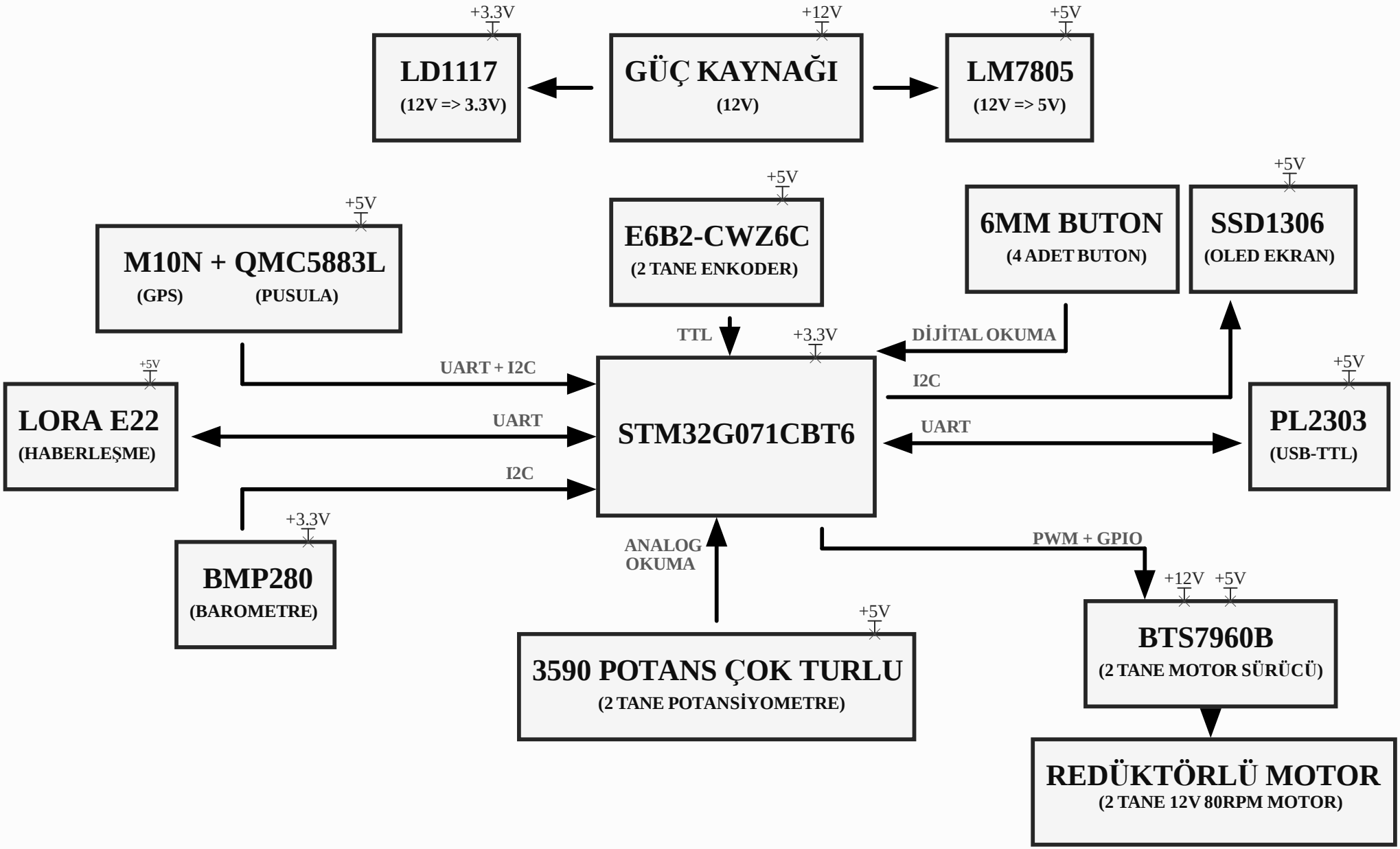
Verici Kartı Blok Diyagramı



Alıcı Kartı Blok Diyagramı

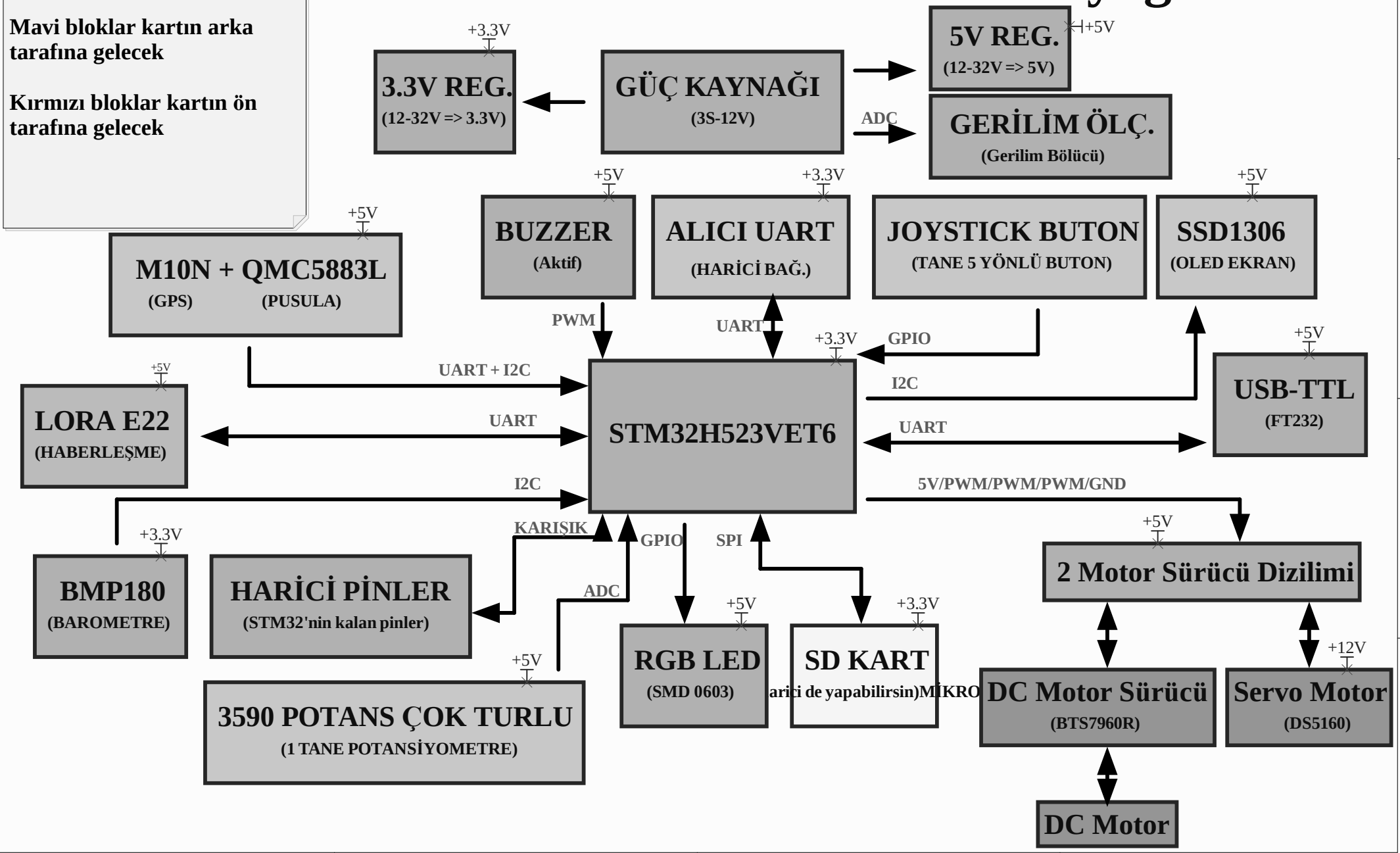


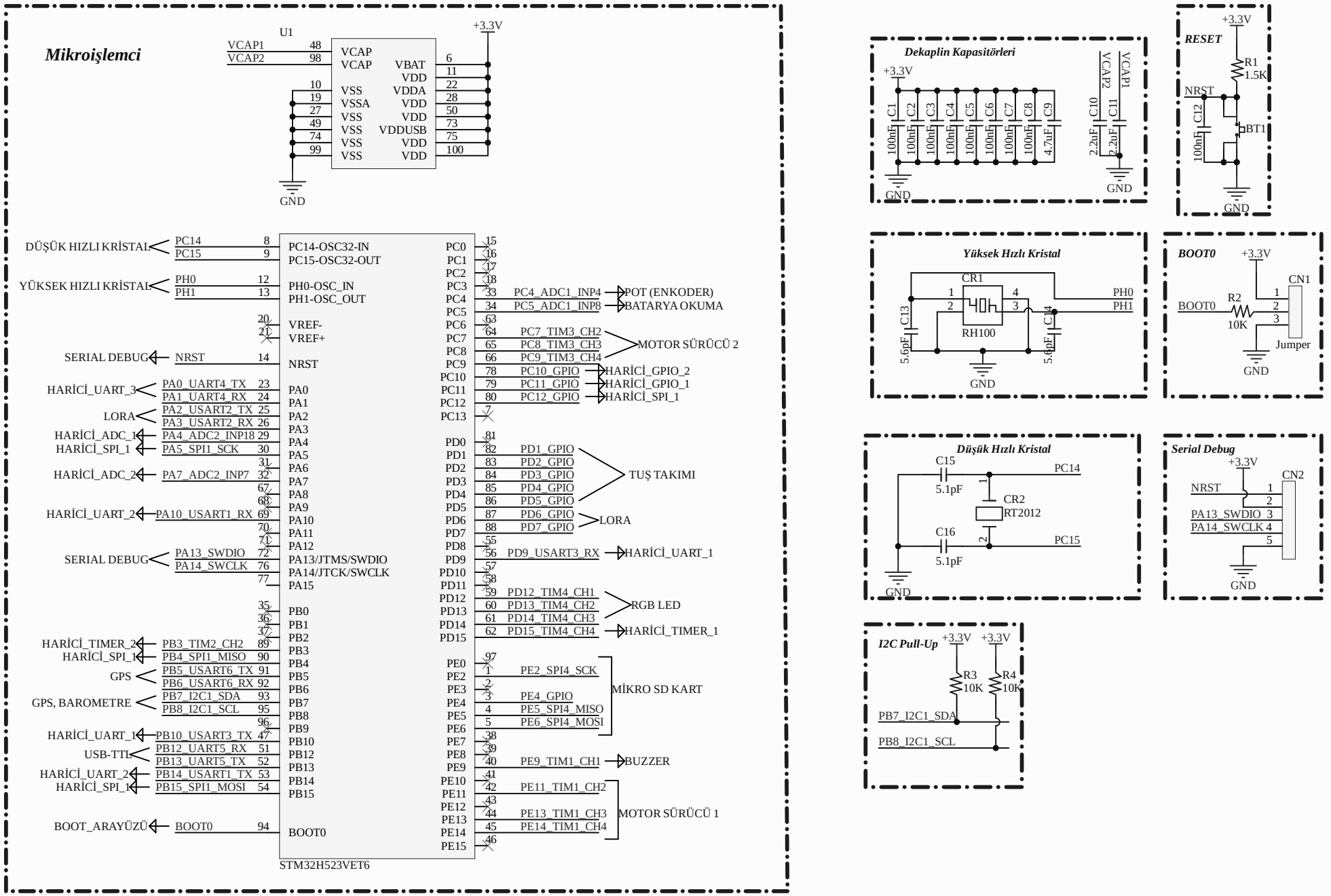
Alıcı Kartı Blok Diyagramı V2

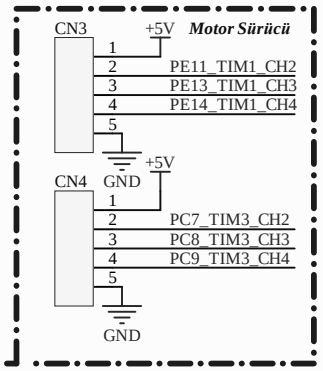
Yeşil bloklar jst ile ayırılacak

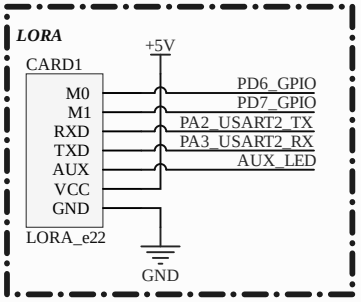
Mavi bloklar kartın arka tarafına gelecek

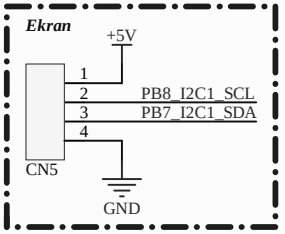
Kırmızı bloklar kartın ön tarafına gelecek

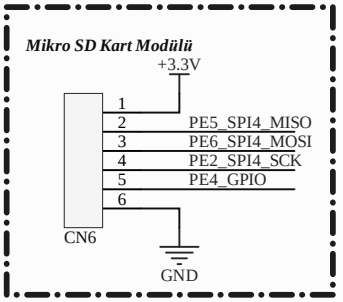


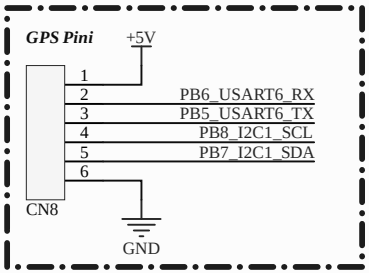

Mikroişlemci
U1
STM32H523VET6
+3.3V
GND
VCAP1
VCAP2
VCAP
VBAT
VDD
VSS
VDDA
VSSA
VDDUSB
PC14-OSC32-IN
PC15-OSC32-OUT
PH0-OSC_IN
PH1-OSC_OUT
VREF-
VREF+
NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13/JTMS/SWDIO
PA14/JTCK/SWCLK
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
BOOT0
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
PE0
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
DÜŞÜK HIZLI KRİSTAL
YÜKSEK HIZLI KRİSTAL
SERIAL DEBUG
HARİCİ_UART_3
LORA
HARİCİ_ADC_1
HARİCİ_SPI_1
HARİCİ_ADC_2
HARİCİ_UART_2
SERIAL DEBUG
HARİCİ_TIMER_2
HARİCİ_SPI_1
GPS
GPS, BAROMETRE
HARİCİ_UART_1
USB-TTL
HARİCİ_UART_2
HARİCİ_SPI_1
BOOT_ARAYÜZÜ
PC4_ADC1_INP4
PC5_ADC1_INP8
PC7_TIM3_CH2
PC8_TIM3_CH3
PC9_TIM3_CH4
PC10_GPIO
PC11_GPIO
PC12_GPIO
PD9_USART3_RX
PD12_TIM4_CH1
PD13_TIM4_CH2
PD14_TIM4_CH3
PD15_TIM4_CH4
PE2_SPI4_SCK
PE4_GPIO
PE5_SPI4_MISO
PE6_SPI4_MOSI
PE9_TIM1_CH1
PE11_TIM1_CH2
PE13_TIM1_CH3
PE14_TIM1_CH4
POT (ENKODER)
BATARYA OKUMA
MOTOR SÜRÜCÜ 2
HARİCİ_GPIO_2
HARİCİ_GPIO_1
HARİCİ_SPI_1
TUŞ TAKIMI
LORA
HARİCİ_UART_1
RGB LED
HARİCİ_TIMER_1
MİKRO SD KART
BUZZER
MOTOR SÜRÜCÜ 1
Dekaplin Kapasitörleri
C1 100nF
C2 100nF
C3 100nF
C4 100nF
C5 100nF
C6 100nF
C7 100nF
C8 100nF
C9 4.7uF
C10 2.2uF
C11 2.2uF
VCAP2
VCAP1
RESET
R1 1.5K
C12 100nF
BT1
NRST
Yüksek Hızlı Kristal
CR1
RH100
C13 5.6pF
C14 5.6pF
PH0
PH1
BOOT0
R2 10K
CN1
Jumper
Düşük Hızlı Kristal
C15 5.1pF
C16 5.1pF
CR2
RT2012
PC14
PC15
Serial Debug
CN2
PA13_SWDIO
PA14_SWCLK
I2C Pull-Up
R3 10K
R4 10K
PB7_I2C1_SDA
PB8_I2C1_SCL

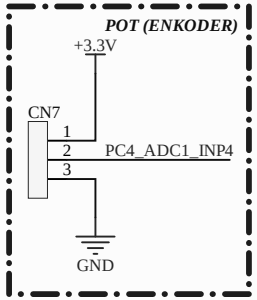

CN3
+5V
Motor Sürücü
1
2 PE11_TIM1_CH2
3 PE13_TIM1_CH3
4 PE14_TIM1_CH4
5
CN4
+5V
GND
1
2 PC7_TIM3_CH2
3 PC8_TIM3_CH3
4 PC9_TIM3_CH4
5
GND

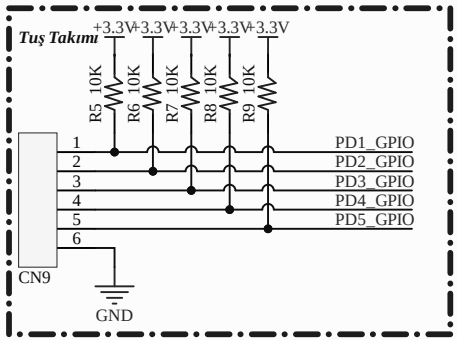

LORA
CARD1
+5V
M0
M1
RXD
TXD
AUX
VCC
GND
PD6_GPIO
PD7_GPIO
PA2_USART2_TX
PA3_USART2_RX
AUX_LED
LORA_e22
GND

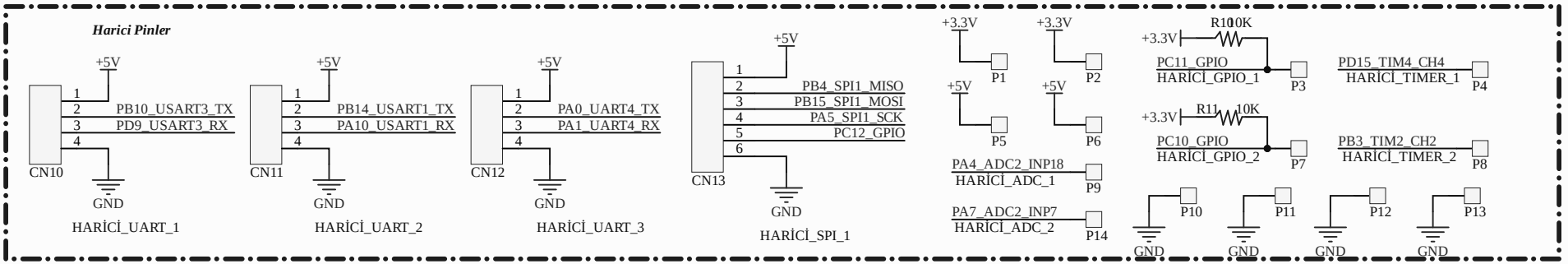

Ekran
+5V
1
2 PB8_I2C1_SCL
3 PB7_I2C1_SDA
4
CN5
GND

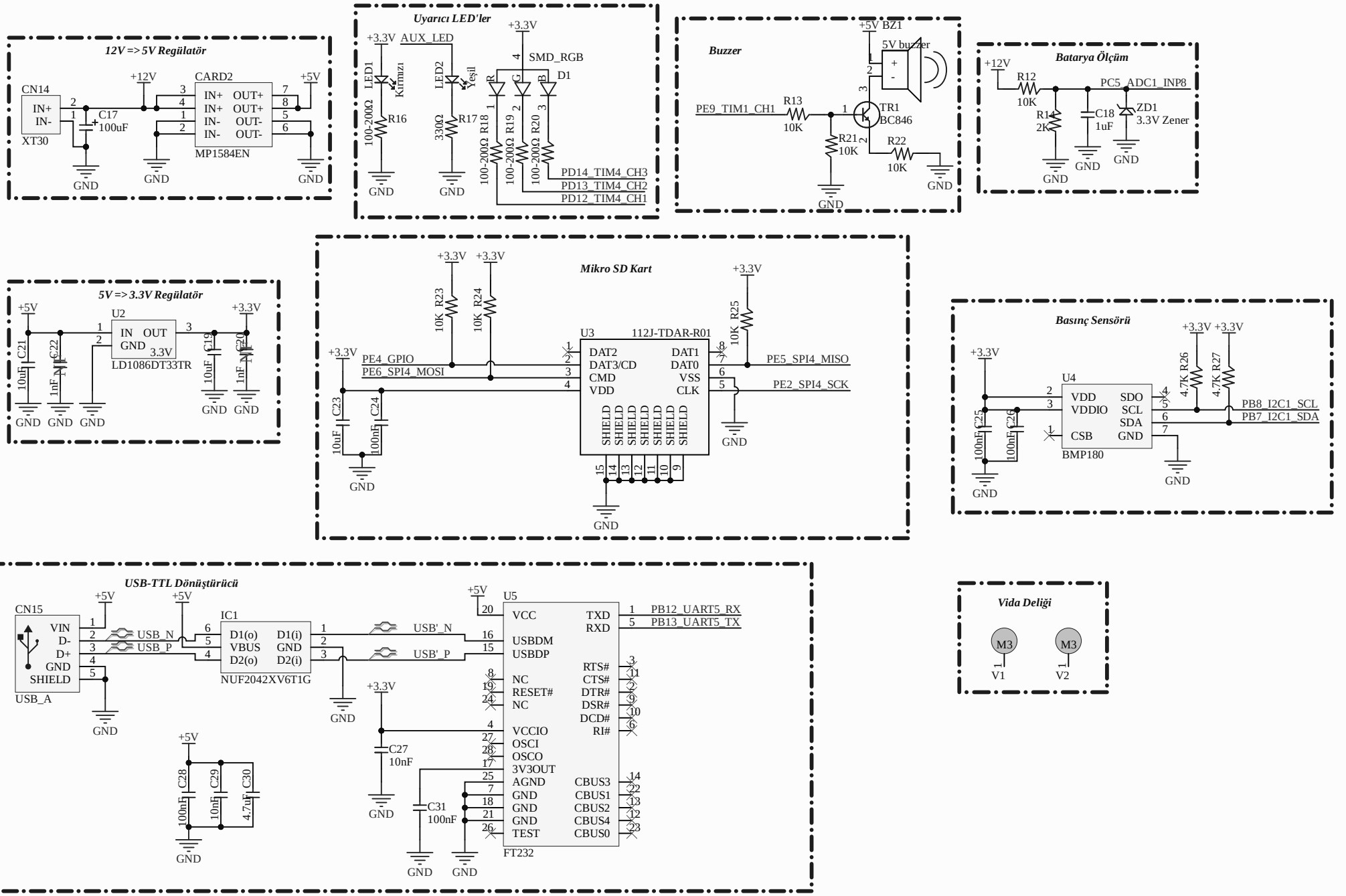

Mikro SD Kart Modülü
+3.3V
1
2 PE5_SPI4_MISO
3 PE6_SPI4_MOSI
4 PE2_SPI4_SCK
5 PE4_GPIO
6
CN6
GND


GPS Pini
+5V
1
2 PB6_USART6_RX
3 PB5_USART6_TX
4 PB8_I2C1_SCL
5 PB7_I2C1_SDA
6
CN8
GND


POT (ENKODER)
+3.3V
CN7
1
2 PC4_ADC1_INP4
3
GND


Tuş Takımı
+3.3V +3.3V +3.3V +3.3V +3.3V
R5 10K
R6 10K
R7 10K
R8 10K
R9 10K
1
2 PD1_GPIO
3 PD2_GPIO
4 PD3_GPIO
5 PD4_GPIO
6 PD5_GPIO
CN9
GND


Harici Pinler
+5V
1
2 PB10_USART3_TX
3 PD9_USART3_RX
4
CN10
GND
HARİCİ_UART_1
+5V
1
2 PB14_USART1_TX
3 PA10_USART1_RX
4
CN11
GND
HARİCİ_UART_2
+5V
1
2 PA0_UART4_TX
3 PA1_UART4_RX
4
CN12
GND
HARİCİ_UART_3
+5V
1
2 PB4_SPI1_MISO
3 PB15_SPI1_MOSI
4 PA5_SPI1_SCK
5 PC12_GPIO
6
CN13
GND
HARİCİ_SPI_1
+3.3V
P1
+3.3V
P2
+5V
P5
+5V
P6
PA4_ADC2_INP18
HARİCİ_ADC_1
P9
PA7_ADC2_INP7
HARİCİ_ADC_2
P14
+3.3V
R10 10K
PC11_GPIO
HARİCİ_GPIO_1
P3
PD15_TIM4_CH4
HARİCİ_TIMER_1
P4
+3.3V
R11 10K
PC10_GPIO
HARİCİ_GPIO_2
P7
PB3_TIM2_CH2
HARİCİ_TIMER_2
P8
P10
GND
P11
GND
P12
GND
P13
GND


12V => 5V Regülatör
CN14
IN+
IN-
XT30
C17
100uF
CARD2
IN+ OUT+
IN+ OUT+
IN- OUT-
IN- OUT-
MP1584EN
+12V
+5V
GND
Uyarıcı LED'ler
+3.3V
AUX_LED
LED1
Kırmızı
LED2
Yeşil
SMD_RGB
D1
R
G
B
100-200Ω
R16
330Ω
R17
R18
R19
R20
PD14_TIM4_CH3
PD13_TIM4_CH2
PD12_TIM4_CH1
Buzzer
+5V BZ1
5V buzzer
PE9_TIM1_CH1
R13
10K
TR1
BC846
R21
10K
R22
10K
Batarya Ölçüm
R12
10K
R14
2K
C18
1uF
ZD1
3.3V Zener
PC5_ADC1_INP8
5V => 3.3V Regülatör
U2
IN OUT
GND 3.3V
LD1086DT33TR
C21
10uF
C22
1nF
C19
10uF
C20
1nF
Mikro SD Kart
U3
112J-TDAR-R01
DAT2
DAT3/CD
CMD
VDD
DAT1
DAT0
VSS
CLK
SHIELD
R23
10K
R24
10K
R25
10K
PE4_GPIO
PE6_SPI4_MOSI
PE5_SPI4_MISO
PE2_SPI4_SCK
C23
10uF
C24
100nF
Basınç Sensörü
U4
VDD
VDDIO
CSB
SDO
SCL
SDA
GND
BMP180
R26
4.7K
R27
4.7K
C25
100nF
C26
100nF
PB8_I2C1_SCL
PB7_I2C1_SDA
USB-TTL Dönüştürücü
CN15
VIN
D-
D+
GND
SHIELD
USB_A
USB_N
USB_P
IC1
D1(o)
VBUS
D2(o)
D1(i)
GND
D2(i)
NUF2042XV6T1G
USB'_N
USB'_P
U5
VCC
USBDM
USBDP
NC
RESET#
NC
VCCIO
OSCI
OSCO
3V3OUT
AGND
GND
GND
GND
TEST
TXD
RXD
RTS#
CTS#
DTR#
DSR#
DCD#
RI#
CBUS3
CBUS1
CBUS2
CBUS4
CBUS0
FT232
PB12_UART5_RX
PB13_UART5_TX
C27
10nF
C28
100nF
C29
10nF
C30
4.7uF
C31
100nF
Vida Deliği
M3
V1
V2